

Notable Side Effects

- Headache, nightmares, or unusual dreams
- Increased alanine aminotransferase
- Upper respiratory infection, urinary infection



Life-Threatening or Dangerous Side Effects

- None reported



Weight Gain

- Reported but not expected



Sedation

- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, take tasimelteon only if planning to have a full night's sleep
- Switch to a different hypnotic

Best Augmenting Agents for Side Effects

- Over-the-counter pain medication for headache
- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 20 mg at bedtime

Dosage Forms

- Capsule 20 mg

How to Dose

- Should be taken at the same time every night
- Should be taken without food



Dosing Tips

- If tasimelteon cannot be taken at its usual time on a given night, the dose for that night should be skipped
- Capsule should be swallowed whole

Overdose

- Limited experience

Long-Term Use

- Not restricted to short-term use
- Continued use is necessary to maintain circadian entrainment

- Has been studied and found safe in a small number of patients for up to 1 year

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Mean elimination half-life 1.3 hours
- Metabolized by CYP450 1A2 and CYP 450 3A4



Drug Interactions

- Inhibitors of CYP450 1A2, such as fluvoxamine, could increase plasma levels of tasimelteon, so combination of tasimelteon with fluvoxamine or other strong CYP450 1A2 inhibitors should be avoided
- Inducers of CYP450 3A4, such as rifampin, could decrease plasma levels of tasimelteon, so combination of tasimelteon with CYP450 3A4 inducers should be avoided
- Efficacy may be reduced if given in conjunction with CYP450 1A2 inducers (e.g., cigarette smoke)

- Inhibitors of CYP450 3A4, such as ketoconazole, could increase plasma levels of tasimelteon

Other Warnings/Precautions

- Tasimelteon should only be administered at bedtime

Do Not Use

- In patients with severe hepatic impairment
- If there is a proven allergy to tasimelteon

Special populations

Renal Impairment

- Dose adjustment not necessary

Hepatic Impairment

- Dose adjustment not necessary for mild to moderate impairment
- Not recommended for patients with severe impairment

Cardiac Impairment

- Not studied

Elderly

- Exposure to tasimelteon is increased by approximately 2-fold



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, administration of tasimelteon during pregnancy resulted in developmental toxicity (embryofetal mortality, neurobehavioral impairment, decreased growth and development in offspring)

Breast Feeding

- Unknown if tasimelteon is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Use caution if patient is breast feeding

The art of psychopharmacology

Potential Advantages

- Those who require long-term treatment
- Circadian rhythm disorders

Potential Disadvantages

- Expensive

Primary Target Symptoms

- Total nighttime sleep time



Pearls

- Tasimelteon acts upon circadian rhythms by stimulating melatonin receptors in the brain's "pacemaker," namely the suprachiasmatic nucleus
- Theoretically, stimulation of melatonin 1 receptors mediates the suppressive effects of melatonin on the suprachiasmatic nucleus
- Theoretically, stimulation of melatonin 2 receptors mediates the phase shifting effect of melatonin
- Tasimelteon may act by promoting the proper entrainment of circadian rhythms underlying a normal sleep-wake cycle; this is supported by biomarker data involving both melatonin and cortisol
- Thus, tasimelteon may also prove effective for treatment of circadian rhythm disturbances such as shift work sleep disorder and

jet lag, and studies in jet lag are currently under way; however, this would be an expensive alternative

- Theoretically could boost the effects of an antidepressant in blind individuals with depression or anxiety, but this has not been properly studied and augmentation studies in depression are not positive
- Lack of actions on GABA systems, which may be related to lack of apparent abuse liability

Suggested Reading

Leger D , Quera-Salva MA , Vecchierini MF , et al. Safety profile of tasimelteon, a melatonin MT1 and MT2 receptor agonist: pooled safety analyses from six clinical studies . *Expert Opin Drug Saf* 2015 ;**14** (11):1673 –85.

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Lockley SW , Dressman MA , Licamele L , et al. Tasimelteon for non-24-hour sleep-wake disorder in totally blind people (SET and RESET): two multicentre, randomised, double-masked, placebo-controlled phase 3 trials . *Lancet* 2015 ;**386** (10005):1754 –64.

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Stahl SM . Mechanism of action of tasimelteon in non-24 sleep-wake syndrome: treatment for a circadian rhythm disorder in blind patients . *CNS Spectr* 2014 ;**19** (6):475 –8 .

Temazepam

Therapeutics

Brands

- Restoril
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

Commonly Prescribed for

(bold for FDA approved)

- **Short-term treatment of insomnia**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour, but can take longer in some patients

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene

- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness
- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth
- Rebound insomnia when withdrawing from long-term treatment



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, only take temazepam if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Switch to a non-benzodiazepine hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 15 mg/day at bedtime

Dosage Forms

- Capsule 7.5 mg, 15 mg, 22 mg, 30 mg

How to Dose

- 15 mg/day at bedtime; may increase to 30 mg/day at bedtime if ineffective



Dosing Tips

- Use lowest possible effective dose and assess need for continued treatment regularly
- Temazepam should generally not be prescribed in quantities greater than a 1-month supply
- Patients with lower body weights may require lower doses
- * Because temazepam is slowly absorbed, administering the dose 1–2 hours before bedtime may improve onset of action and shorter sleep latency
- Risk of dependence may increase with dose and duration of treatment

Overdose

- Can be fatal in monotherapy; slurred speech, poor coordination, respiratory depression, sedation, confusion, coma

Long-Term Use

- Not generally intended for long-term use

Habit Forming

- Temazepam is a Schedule IV drug
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses

- History of drug addiction may increase risk of dependence

How to Stop

- If taken for more than a few weeks, taper to reduce chances of withdrawal effects
- Patients with history of seizure may seize upon sudden withdrawal
- Rebound insomnia may occur the first 1–2 nights after stopping
- For patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

Pharmacokinetics

- No active metabolites
- Half-life approximately 8–15 hours



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)
- If temazepam is used with kava, clearance of either drug may be affected

Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Temazepam should only be administered at bedtime

Do Not Use

- If patient is pregnant
- If patient has angle-closure glaucoma
- If there is a proven allergy to temazepam or any benzodiazepine

Special populations

Renal Impairment

- Recommended dose: 7.5 mg/day

Hepatic Impairment

- Recommended dose: 7.5 mg/day

Cardiac Impairment

- Dosage adjustment may not be necessary
- Benzodiazepines have been used to treat insomnia associated with acute myocardial infarction

Elderly

- Recommended dose: 7.5 mg/day



Children and Adolescents

- Safety and efficacy have not been established

- Long-term effects of temazepam in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored

Pregnancy

- Contraindicated for use in pregnancy
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy

Breast Feeding

- Unknown if temazepam is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss

The art of psychopharmacology

Potential Advantages

- Patients with middle insomnia (nocturnal awakening)

Potential Disadvantages

- Patients with early insomnia (problems falling asleep)

Primary Target Symptoms

- Time to sleep onset
- Total sleep time
- Nighttime awakenings

Pearls

- If tolerance develops, it may result in increased anxiety during the day and/or increased wakefulness during the latter part of the night
- * Slow gastrointestinal absorption compared to other sedative benzodiazepines, so may be more effective for nocturnal awakening than for initial insomnia unless dosed 1–2 hours prior to bedtime
- * Notable for delayed onset of action compared to some other sedative hypnotics
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Ashton H . Guidelines for the rational use of benzodiazepines . When and what to use. *Drugs* 1994 ;**48** :25 –40 .

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Fraschini F , Stankov B . Temazepam: pharmacological profile of a benzodiazepine and new trends in its clinical application . *Pharmacol Res* 1993 ;**27** :97 –113.

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Heel RC , Brogden RN , Speight TM , Avery GS . Temazepam: a review of its pharmacological properties and therapeutic efficacy as an hypnotic . *Drugs* 1981 ;**21** :321–40 .

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McElnay JC , Jones ME , Alexander B . Temazepam (Restoril, Sandoz Pharmaceuticals) . *Drug Intell Clin Pharm* 1982 ;**16** :650 –6 .

Thioridazine

Therapeutics

Brands

- Mellaril
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenic patients who fail to respond to treatment with other antipsychotic drugs**

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Is a second-line treatment option
- * Should evaluate for switching to an antipsychotic with a better risk/benefit ratio

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Augmentation of thioridazine has not been systematically studied and can be dangerous, especially with drugs that can either prolong QTc interval or raise thioridazine plasma levels

Tests

- * Baseline ECG and serum potassium levels should be determined
- * Periodic evaluation of ECG and serum potassium levels
- Serum magnesium levels may also need to be monitored
- * Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Phenothiazines may cause false-positive phenylketonuria results
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and thioridazine should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can

cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)

- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
 - Antihistaminic actions may cause sedation, weight gain
 - Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
 - Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown
- * Mechanism of potentially dangerous QTc prolongation may be related to actions at ion channels

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia
- * Priapism
- * Drug-induced parkinsonism
- * Tardive dyskinesia
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea

✱ Pigmentary retinopathy at high doses

- Dizziness, sedation
- Dry mouth, constipation, blurred vision
- Decreased sweating
- Sexual dysfunction
- Hypotension
- Weight gain



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
 - Rare jaundice, agranulocytosis
 - Rare seizures
- ✱ Dose-dependent QTc prolongation
- Ventricular arrhythmias and sudden death
 - As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain

